Eine Woche, ein Beispiel

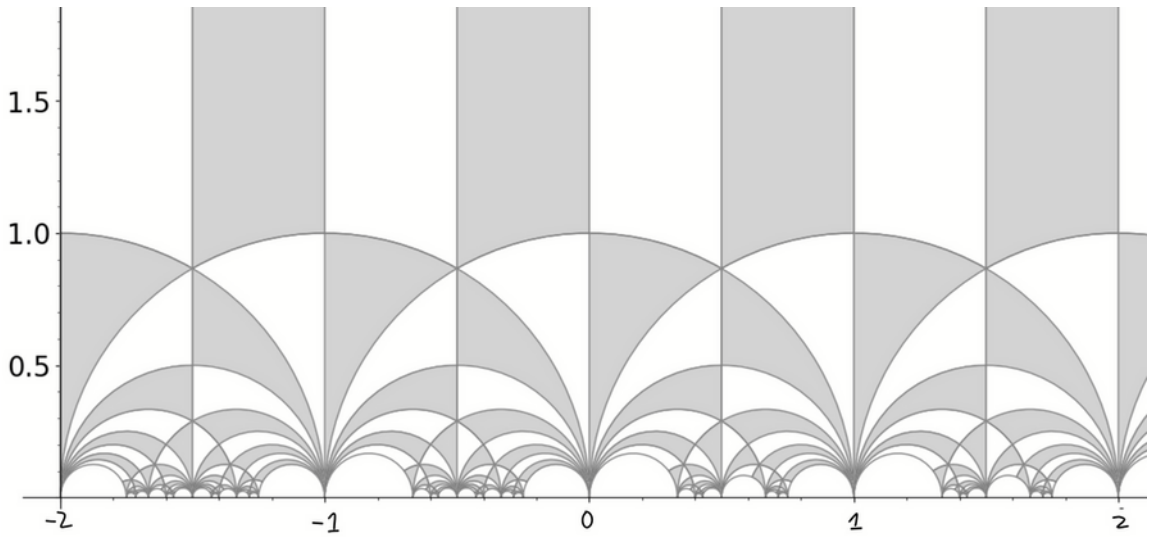
12.28 binary quadratic form

This note should be inserted in [2025.12.21], Section 2.
In fact, the geometry of binary quadratic forms is a bit interesting, regardless of its relation with Hodge structures.

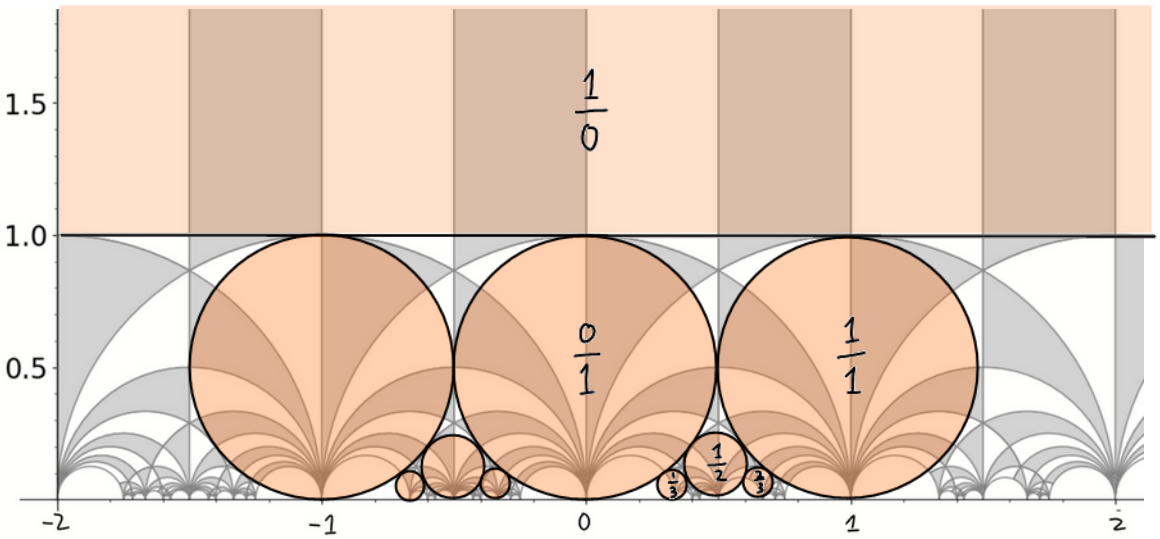
1 several figures in $\mathcal{H}$
2. equivalent expressions of $M_{sym}^{2\times 2}(\mathbb{Z})$
3. some invariants

1 several figures in $\mathcal{H}$ [JC97, Chapter 1]

①. Fundamental domain of $SL_2(\mathbb{Z})$



②. Ford circles

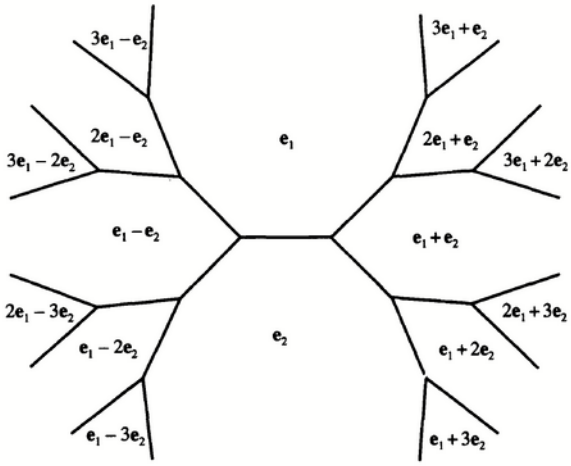


$$\mathbb{Q} \cup \{\infty\} = \text{“}\mathbb{P}\mathbb{Z}^2 = (\mathbb{Z}^2 - \{0\}) / (\mathbb{Z} - \{0\})\text{”} = SL_2(\mathbb{Z}) / \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$$

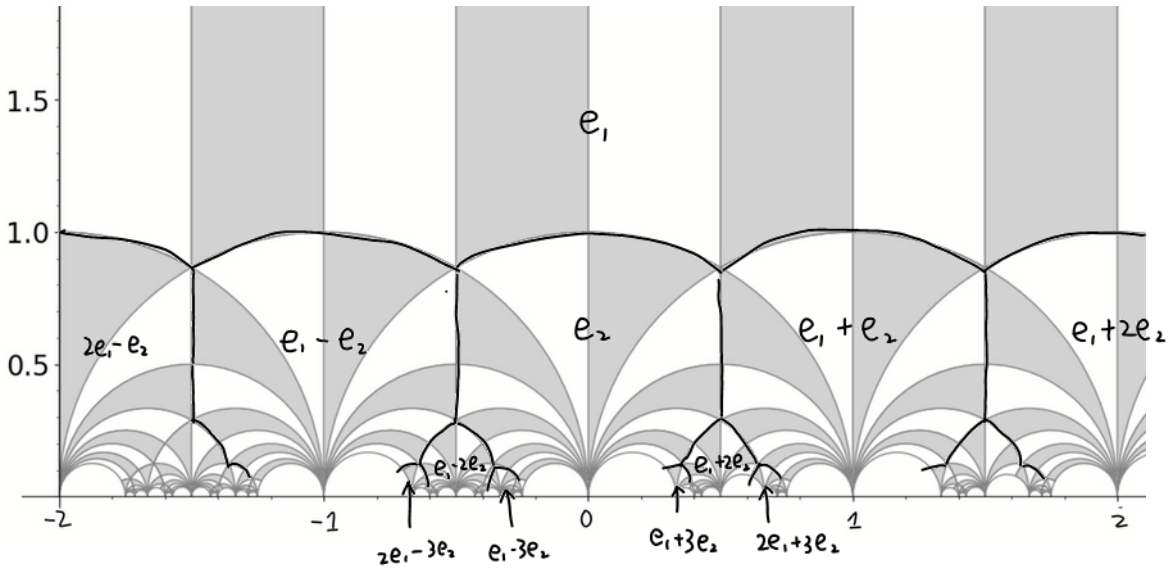
The "distance" between two circles $\frac{p_1}{q_1}$ & $\frac{p_2}{q_2}$ is not $\begin{vmatrix} p_1 & p_2 \\ q_1 & q_2 \end{vmatrix}$

How to compute it?

③ Conway's topograph



[JC97, p8]



$$\mathbb{Z}^2_{prim}/\{\pm 1\} \cong \mathbb{Q} \cup \{\infty\}$$

Ford circle related with SL_2(Z)-action:
https://mathoverflow.net/questions/25676/canonical-fundamental-domain-for-a-discrete-subgroup-%ce%93-of-sl%e2%82%82r-acting-on-hyper
I thought that the afterthoughts in [JC97] is more useful.

perpendicular bisector:
https://math.stackexchange.com/questions/2220933/perpendicular-bisector-of-a-hyperbolic-line
https://www.researchgate.net/publication/221435311_Hyperbolic_Voronoi_Diagram (Lemma 2,3 gives what I want)

geometrical intuition of hyperbolic circles:
https://www.youtube.com/watch?v=mk2et2_2zzk